D07 · Warehouse Automation & AMR/Robotics

04 · Conveyor systems, sorters & warehouse control systems (WCS)

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Conveyors move cases and totes between zones, sorters divert them to destinations at high speed, and the Warehouse Control System (WCS) coordinates this equipment in real time, translating WMS instructions into machine commands.

2 WHY IT MATTERS

At high volume, conveyors and sorters provide a throughput and accuracy that manual sorting cannot. **They are only as good as the data that tells them where each item goes**: unreadable labels and missing master data end up in the reject lane. Because a stopped sorter at peak halts the whole flow, maintenance and spare parts are part of the investment.

3 HOW IT IS DONE

1. **Confirm volume justifies it.** Sustained daily volume and number of destinations high enough for payback.
2. **Design WMS–WCS integration upfront.** Routing logic, exceptions and status messages defined before build.
3. **Plan maintenance and spares.** Preventive schedules, critical spares and response-time agreements.
4. **Monitor the reject lane.** Reject rate is the early warning for label and data quality problems.

IN PRACTICE

Sorters only have ROI with sufficiently high volumes — don’t underestimate requirements

A mid-capacity crossbelt sorter costs $2–5M. ROI is only achieved with >20,000 items/day and many classification destinations.

Preventive sorter maintenance is the most important investment for availability

A sorter that stops at peak volume has the highest possible operational impact.

The WCS must be integrated with the WMS for correct routing of each box

Without WMS-WCS integration, the sorter doesn’t know where to send each box because it doesn’t know the box content.

Monitor the sorter reject bucket — it’s the indicator of data quality problems

Boxes going to the reject bucket are those the system couldn’t identify. A high reject bucket indicates a labeling or master data problem.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Crossbelt sorter — FMCG distribution DC, 35,000 boxes/day — Current sorting process is manual with 80 operators.

Metric	Manual sorting	Crossbelt sorter
Sorting capacity (boxes/hour)	HIGH	3,200 boxes/hour with 80 operators
Sorter capacity	OK	9,600 boxes/hour with 3 operators (1 supervision + 2 feeding)
Sorting accuracy (box to correct destination)	OK	97.8% manual (1,540 errors/day) -> 99.99% (0.35 errors/day)

Result: Sorter implemented: Capacity +200%. Sorting operators: 80 -> 3. ROI at 3 years: $3.6M labor savings + $1.2M error reduction vs. $4.2M investment. Payback: 2.6 years.

HOW IT IS MEASURED

Sorter Throughput (boxes sorted per hour)

Sorter ThroughputTotal boxes processed by the sorter / Operating hours

Benchmark: 5,000–15,000 boxes/hour depending on type · Real throughput must be >80% of manufacturer nominal Real throughput <70% of nominal indicates feeding problems, code reading problems, or mechanical issues.

Sorter Accuracy %

Sorter Accuracy %(Boxes sent to correct destination / Total boxes processed by sorter) × 100

Benchmark: >99.9% accuracy for correctly calibrated sorters Sorter accuracy <99.5% indicates barcode reading problems that generate boxes to the reject bucket or incorrect destination.

WHAT YOU WOULD USE

Conveyor & Sorter

Interroll / Vanderlande — Module: Conveyor Systems + Crossbelt SortersThe global reference conveyor and sorter system providers. Vanderlande is part of Toyota Industries.

Dematic / SSI Schaefer — Module: WCS + Integrated AutomationProviders offering both automation hardware and integrated WCS.

4 SO WHAT

Automate flow where volume is sustained, and protect it. Justify on volume, integrate WMS and WCS carefully, fund maintenance and watch the reject lane as a data-quality gauge.

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